

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1653

Parkin 138 kV
February 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1653. The Interconnection Customer proposes a 213.4 MW solar plus storage facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on January 23, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 374 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,482,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Alderwood - Waynesboro 138 KV	Bear Substation - Clover 345 KV	108.1	377	6.6
Grassmere - Coal Mtn 138 KV L	Bear Substation - Clover Substation 345 KV line	108.1	329	6.6
Central Austin - Waynesboro 138 KV	Bear - Clover 345 KV	108.1	348	6.6
Central Austin - Coal Mtn 138 KV	Bear - Cover Substation 345 KV line	108.1	338	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alderwood - Waynesboro 138 KV	Reconductor line	\$1,895,000	20% (\$389,000)
Grassmere - Coal Mtn 138 KV L	Reconductor line	\$1,072,000	20% (\$220,000)
Central Austin Substation - Waynesboro 138 KV	Reconductor line	\$1,826,000	20% (\$374,000)
Central Austin - Coal Mtn 138 KV	Reconductor line	\$2,431,000	20% (\$499,000)
	Total allocated to project		\$1,482,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.